

SynDiff Synthetic Benchmark — Live Progress

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$S = I(X_1; Y | X_2) - I(X_1; Y)$ in nats; $S > 0$ synergy-dominant, $S < 0$ redundancy-dominant. SynDiff (ours) vs prior estimators.

Dataset construction

Phase 1 — Gaussian (closed-form ground truth)

Scalar base: $X_1, X_2 \sim \mathcal{N}(0, 1)$ with $\text{corr}(X_1, X_2) = \rho$; label $Y = X_1 + X_2 + \sigma_y \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$. Thus $\text{Var}(Y) = 2 + 2\rho + \sigma_y^2$ and $\text{Cov}(X_i, Y) = 1 + \rho$. To lift to d dimensions the informative scalar is padded with $d - 1$ i.i.d. $\mathcal{N}(0, 1)$ coordinates and rotated by a fixed per-seed orthogonal matrix Q_i (one per variable). Rotation and independent padding preserve mutual information, so S is unchanged and the closed form carries over. Ground truth is computed from the 3×3 covariance via $I = \frac{1}{2} \log(\det \Sigma_A \det \Sigma_B / \det \Sigma_{AB})$ and the analogous conditional form. Cells sweep $\rho \in \{-.9, \dots, .9\}$, $\sigma_y \in \{.1, 1\}$, $n \in \{1k, \dots, 100k\}$, $d \in \{1, 8, 32, 128\}$.

Phase 2 — discrete U/R/S (mixture-density MC truth)

Latent sources $s_1, s_2, s_3 \sim \text{Unif}\{0, \dots, Q - 1\}$ with $Q = 8$. Encodings $X_i = W_i \text{onehot}(s_i) + \sigma_{\text{enc}} \mathcal{N}(0, I_{32})$ with orthonormal $W_i \in \mathbb{R}^{32 \times 8}$ frozen per seed. Regimes (X_1 always encodes s_1): **UNIQUE** $Y = s_1$; **REDUNDANT** $Y = s_1$ and X_2 also encodes s_1 ; **SYNERGY1** $Y = (s_1 + s_2) \bmod Q$ with $X_2 \sim s_2$; **SYNERGY2** $Y = (s_3 + (s_1 + s_2) \bmod Q) \bmod Q$ with $X_2 = [\text{enc}(s_2) | \text{enc}(s_3)] \in \mathbb{R}^{64}$; **INDEPENDENT** $Y = s_3$. The λ -dial mixes the SYNERGY1 and REDUNDANT rules per sample ($\text{Bernoulli}(\lambda)$). Ground truth is the exact $S = I(X_1; X_2, Y) - I(X_1; X_2) - I(X_1; Y)$ obtained by Monte Carlo over the enumerable Gaussian-mixture components (matches the discrete limit as $\sigma_{\text{enc}} \rightarrow 0$: $\pm \log Q$ for pure synergy/redundancy, 0 for unique/independent).

Method: SynDiff

Target $S = I(X_1; Y | X_2) - I(X_1; Y)$ in nats. By the symmetry of interaction information this equals $I(X_1; X_2 | Y) - I(X_1; X_2)$, which the four-mask form below estimates directly. **VP schedule**: $\beta(t) = 0.1 + 19.9t$, $B(t) = \int_0^t \beta = 0.1t + 9.95t^2$, signal $k(t) = e^{-B/2}$, noise $\sigma^2(t) = 1 - e^{-B}$, weight $w(t) = \beta(t)/(2\sigma^2(t))$. A single noise-prediction network $\hat{e}(x_{1,t}, t, \text{cond})$ (*MaskedCondMLP*: Fourier- t features $\oplus$ per-slot conditioning embeddings each with a learned NULL token; 4×512 SiLU) is trained to denoise X_1 while each conditioning slot (X_2, Y) is independently dropped to its NULL token with probability 0.15. EMA weights (decay 0.999) are used at evaluation. **Four-mask estimator**. Per evaluation triple and per stratified time t ($K=16$ strata on $[10^{-3}, 1]$) with one shared noise draw ε and $x_{1,t} = k(t)x_1 + \sigma(t)\varepsilon$:

$$\begin{aligned} e_1 &= \hat{e}(x_{1,t}, t, x_2, y), & e_2 &= \hat{e}(x_{1,t}, t, \emptyset, y), \\ e_3 &= \hat{e}(x_{1,t}, t, x_2, \emptyset), & e_4 &= \hat{e}(x_{1,t}, t, \emptyset, \emptyset), \end{aligned}$$

$$\text{integrand}(t) = w(t)(\|e_1 - e_2\|^2 - \|e_3 - e_4\|^2).$$

Per-sample $\hat{S} = (1 - 10^{-3}) \overline{\text{integrand}_t}$; the estimate is the mean over 10^4 fresh triples with a 1000-sample bootstrap CI (norms in fp32, no-grad, EMA weights). **Baselines**: *minde_comp* (three

separately trained MINDE MI terms composed as above), *MINE* and *InfoNCE* (lower-bound MI, same composition), *batch_pid* (Liang et al. CVX PID, k-means $k=20$ then max-entropy solve), and *broja/plugin* on $d=1$.

Sweep progress

Phase	Method	Done	Expected
P1	batch_pid	32	96
P1	broja	0	6
P1	infonce	96	96
P1	minde_comp	96	96
P1	mine	96	96
P1	plugin	2	6
P1	syndiff	96	96
P2	batch_pid	30	90
P2	infonce	53	90
P2	minde_comp	50	90
P2	mine	54	90
P2	syndiff	53	90

Accuracy vs ground truth (completed cells, lower error better)

Method	#cells	mean $ \text{est} - \text{truth} $	sign acc
minde_comp	49	0.1195	0.80
plugin	2	0.1343	1.00
syndiff	50	0.2542	0.88
batch_pid	62	0.3669	0.74
mine	50	0.7918	0.60
infonce	50	4.2589	0.40

SynDiff vs prior — head to head (nats)

Cell	truth	SynDiff	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rho0_sy0p1_d8_n20k	+1.963	+1.887	+0.147	+1.922	+0.631	-5.604	—
p1_dial_rho0_sy1_d8_n20k	+0.144	+0.136	+0.154	+0.145	-0.555	-5.315	—
p1_dial_rho0p3_sy0p1_d8_n20k	+1.740	+1.677	+0.128	+1.678	+0.627	-5.460	—
p1_dial_rho0p3_sy1_d8_n20k	+0.007	+0.006	+0.129	+0.010	-0.603	-5.429	—
p1_dial_rho0p5_sy0p1_d8_n20k	+1.477	+1.407	+0.115	+1.434	+0.525	-5.436	—
p1_dial_rho0p5_sy1_d8_n20k	-0.134	-0.134	+0.108	-0.128	-0.829	-5.254	—
p1_dial_rho0p7_sy0p1_d8_n20k	+1.035	+0.971	+0.090	+0.991	+0.309	-5.395	—
p1_dial_rho0p7_sy1_d8_n20k	-0.329	-0.334	+0.098	-0.321	-0.651	-5.190	—
p1_dial_rho0p9_sy0p1_d8_n20k	+0.024	-0.032	+0.055	-0.025	-0.329	-5.321	—
p1_dial_rho0p9_sy1_d8_n20k	-0.610	-0.601	+0.066	-0.612	-0.807	-5.145	—
p1_dial_rho0p3_sy0p1_d8_n20k	+2.047	+1.976	+0.161	+2.017	+0.724	-5.385	—
p1_dial_rho0p3_sy1_d8_n20k	+0.209	+0.199	+0.157	+0.210	-0.562	-5.363	—
p1_dial_rho0p5_sy0p1_d8_n20k	+2.023	+1.953	+0.149	+1.980	+0.795	-5.507	—

Cell	truth	SynDiff	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rhon0p5_sy1_d8_n20k	+0.213	+0.197	+0.153	+0.214	-0.504	-5.342	—
p1_dial_rhon0p7_sy0p1_d8_n20k	+1.896	+1.826	+0.138	+1.853	+0.689	-5.488	—
p1_dial_rhon0p7_sy1_d8_n20k	+0.177	+0.158	+0.148	+0.182	-0.648	-5.374	—
p1_dial_rhon0p9_sy0p1_d8_n20k	+1.473	+1.412	+0.142	+1.445	+0.466	-5.470	—
p1_dial_rhon0p9_sy1_d8_n20k	+0.083	+0.054	+0.142	+0.089	-0.580	-5.200	—
p1_dim_rho0p7_sy1_d128_n20k	-0.329	-0.050	+0.135	-0.295	-1.852	-6.133	—
p1_dim_rho0p7_sy1_d1_n20k	-0.329	-0.329	-0.240	-0.322	-0.321	-0.271	-0.191
p1_dim_rho0p7_sy1_d32_n20k	-0.329	-0.365	+0.133	-0.330	-1.467	-6.102	—
p1_dim_rhon0p7_sy1_d128_n20k	+0.177	+0.022	+0.136	+0.271	-1.938	-6.143	—
p1_dim_rhon0p7_sy1_d1_n20k	+0.177	+0.168	+0.239	+0.184	+0.183	+0.171	+0.308
p1_dim_rhon0p7_sy1_d32_n20k	+0.177	+0.129	+0.137	+0.177	-1.085	-6.105	—
p1_ns_rho0p7_sy1_d8_n100k	-0.329	-0.332	+0.098	-0.321	-0.704	-5.204	—
p1_ns_rho0p7_sy1_d8_n1k	-0.329	-0.331	+0.098	-0.325	-0.630	-5.217	—
p1_ns_rho0p7_sy1_d8_n20k	-0.329	-0.333	+0.098	-0.322	-0.652	-5.261	—
p1_ns_rho0p7_sy1_d8_n5k	-0.329	-0.331	+0.098	-0.321	-0.651	-5.174	—
p1_ns_rhon0p7_sy1_d8_n100k	+0.177	+0.157	+0.149	+0.180	-0.648	-5.374	—
p1_ns_rhon0p7_sy1_d8_n1k	+0.177	+0.160	+0.148	+0.183	-0.497	-5.338	—
p1_ns_rhon0p7_sy1_d8_n20k	+0.177	+0.154	+0.148	+0.180	-0.602	-5.375	—
p1_ns_rhon0p7_sy1_d8_n5k	+0.177	+0.154	+0.148	+0.185	-0.453	-5.449	—
p2_independent_senc0p1	+0.000	+0.000	+0.136	-0.001	-0.305	-1.923	—
p2_independent_senc0p5	+0.000	+0.000	+0.141	-0.000	-0.763	-2.083	—
p2_independent_senc1	+0.000	-0.000	+0.136	-0.000	-0.844	-2.070	—
p2_lambda0_senc0p1	-2.079	-6.167	-2.068	-2.590	-2.063	-2.191	—
p2_lambda0_senc0p5	-0.730	-0.488	-0.212	-0.645	-1.009	-2.022	—
p2_lambda0_senc1	-0.065	-0.000	+0.144	—	-0.975	-2.077	—
p2_redundant_senc0p1	-2.079	-5.757	-2.062	-3.164	-2.058	-2.194	—
p2_redundant_senc0p5	-0.730	-0.425	-0.216	-0.635	-0.924	-2.035	—
p2_redundant_senc1	-0.065	-0.000	+0.133	-0.041	-0.768	-2.071	—
p2_synergy1_senc0p1	+2.079	+2.252	+2.067	+2.806	+0.824	-2.034	—
p2_synergy1_senc0p5	+0.733	+0.546	+0.461	+0.608	-0.918	-2.094	—
p2_synergy1_senc1	+0.067	-0.000	+0.145	-0.000	-1.015	-2.055	—
p2_synergy2_senc0p1	+2.079	+0.000	+0.719	-0.001	+0.623	-1.968	—
p2_synergy2_senc0p5	+0.424	-0.000	+0.148	-0.000	-0.870	-2.084	—
p2_synergy2_senc1	+0.010	+0.000	+0.146	+0.000	-0.748	-2.064	—
p2_unique_senc0p1	-0.000	-0.000	+0.007	+0.009	-0.838	-2.042	—
p2_unique_senc0p5	-0.000	+0.000	+0.140	+0.005	-1.128	-2.075	—
p2_unique_senc1	-0.000	+0.000	+0.148	+0.002	-1.070	-2.071	—